

API Documentation

Base URL : `http://localhost:8000/api`

All requests and response bodies use - Content-Type : `application/json`

Endpoints

1. **/shorten/ URL:**

POST `/shorten/`

-> this creates a new shortened url for every long (original) url inputted by the user. If the user enters the same `original_url` which has already been shortened before, then the existing record is returned.

The endpoint is rate-limited to 5 requests per minute per IP address

Request Body:

```
{
  "original_url": "https://www.example.com/some/long/url_path"
}
```

Response - 201 : New URL is created

```
{
  "id": 1,
  "original_url": "https://www.example.com/some/very/long/url_path",
  "short_url": "aB3xYz",
  "created_at": "2026-05-17T10:00:00.000000Z"
}
```

Response - 200 : URL already exists

```
{
  "id": 1,
  "original_url": "https://www.example.com/some/very/long/path?query=value",
  "short_url": "aB3xYz",
  "created_at": "2025-05-17T09:45:00.000000Z"
}
```

Response - 400 : Bad Request

```
{
  "original_url": ["Enter a valid URL."]
}
```

Response - 429 : Too Many Requests (rate limit exceeded)

```
{
  "error": "Rate limit exceeded",
  "retry_after_seconds": 34
}
```

Response 503 : Service Unavailable

```
{
  "error": "Failed to generate a unique short URL after multiple attempts."
}
```

2. List all shortened URLs

GET /shorten/

- > returns all shortened URLs ordered by creation date (newest first), each with its total click count

Request body - none

Response - 200 : OK

```
[
  {
    "id": 2,
    "original_url": "https://www.another-example.com",
    "short_url": "mN7pQr",
    "created_at": "2025-05-17T11:00:00.000000Z",
    "clicks": 14
  },
  {
    "id": 1,
    "original_url": "https://www.example.com/some/very/long/path?query=value",
    "short_url": "aB3xYz",
    "created_at": "2025-05-17T10:00:00.000000Z",
    "clicks": 3
  }
]
```

3. Track a click

POST /analytics/{short_url_code}/click/

-> records a click event for a given short url and returns the updated total click count.
Used by the analytics page for analysis of the url.

Request body - none

Response - 201 created

```
{  
  "clicks": 4  
}
```

Response - 404 : not found

```
{  
  "error": "Short URL not found."  
}
```

4. Get url analysis

GET /analytics/{short_url_code}/

-> returns click analytics for a specific shortened url, including total click and a day-to-day breakdown for the last 7 days. Used to populate chart.js line charts.

Request body - none

Response - 200 : OK

```
{  
  "short_url": "aB3xYz",  
  "original_url": "https://www.example.com/some/very/long/path?query=value",  
  "total_clicks": 17,  
  "timeline": [  
    { "date": "2025-05-11", "count": 2 },  
    { "date": "2025-05-13", "count": 5 },  
    { "date": "2025-05-14", "count": 3 },  
    { "date": "2025-05-15", "count": 4 },  
    { "date": "2025-05-17", "count": 3 }  
  ]  
}
```

// only days that have at least one click appear in the timeline. Days with 0 clicks are omitted from the response. The frontend accounts for this when chart labels are built.

Response - 404 : Not Found

```
{  
  "error": "Short URL not found."  
}
```

HTTP Status Code Reference:

Code	Meaning
200 OK	Request succeeded, resource already existed
201 Created	201 Created
400 Bad Request	Invalid input (e.g. malformed URL)
404 Not Found	The requested short URL does not exist
429 Too Many Requests	Rate limit exceeded — check retry_after_seconds
503 Service Unavailable	Server-side error during alias generation